

Systematic Credit Project

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May 2026

1 Overview

I built a reproducible, end-to-end pipeline for a simple systematic credit strategy using U.S. High Yield bond data and macroeconomic indicators. The goal was not to maximize performance, but to build a clean, modular workflow that could realistically be extended into a production setting.

The pipeline covers data ingestion, feature engineering, macro regime classification, bond-level signal construction, portfolio generation, and backtesting. Each component is modular and driven by configuration, allowing the full process to be reproduced with a single run.

2 Signal Design

The signal ranks bonds based on compensation relative to risk:

$$\text{signal} = \text{Yield-to-Worst} + \text{OAS} - \text{DTS} - \text{Duration} - \text{CCC Penalty}$$

The intuition is to favor bonds that offer higher yield and spread, while penalizing those that take on excessive credit risk (DTS), interest rate risk (duration), and very low credit quality (CCC). This is a simple carry vs. risk framework rather than a predictive return model.

3 Macro Regime

To incorporate macro-specific context, I construct a regime score using unemployment, VIX, a derived HY OAS proxy, and the slope of the yield curve. Higher unemployment, higher VIX, and wider spreads indicate more risk, while a steeper yield curve indicates less risk.

Each series is normalized using a 36-month rolling window, which balances stability and responsiveness, and is treated as a configurable parameter. The resulting score classifies periods as risk-on, neutral, or risk-off. The first 36 months are treated as a warm-up period.

4 Portfolio Construction

Each month, I construct a long-only portfolio by selecting the top percentile of bonds ranked by the signal. I use a percentile approach rather than a fixed number of bonds because the universe size changes over time, which keeps selectivity consistent.

The portfolio includes issuer caps, sector caps, and exposure limits for CCC-rated bonds. Given that the sample HY universe is heavily concentrated in Industrials, I focus more on controlling issuer and credit risk rather than forcing strict sector diversification.

5 Backtest and Results

The strategy is evaluated using next-month excess returns. I track standard metrics including return, volatility, Sharpe ratio, drawdown, and turnover.

The Sharpe ratio is modest, which is expected given the simplicity of the signal and the fact that it ranks carry vs. risk rather than forecasting returns. The framework is designed to be extensible and more sophisticated signals or weighting schemes would likely improve performance.

6 Failure Modes

There are several ways this approach could break in practice:

- **Macro regime lag:** The use of rolling normalization means that the regime signal reacts with a delay, which can lead to incorrect classification during rapid market shifts.
- **Liquidity and turnover:** The strategy may require frequent rebalancing, and some bonds in the high yield universe may not be liquid enough to support this level of trading.
- **Data quality:** Missing, stale, or incorrect bond data (e.g., OAS or returns) can distort the signal and lead to poor portfolio decisions.
- **Concentration risk:** Despite constraints, the portfolio may still become overly exposed to certain issuers or lower-quality credit, especially given the structure of the sample high yield universe.

These risks can be monitored through turnover, exposure, and data validation diagnostics.

7 Potential Future Improvements

The next steps would include replacing the internal OAS proxy with benchmark index data, introducing signal weighting, incorporating transaction costs, and adding regime-based exposure scaling.

8 Conclusion

This project demonstrates a structured and reproducible approach to systematic credit research, with a focus on clarity, robustness, and extensibility.